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Breakthrough Low Temperature Dissolvable Rubbers for Dissolvable Plug and Field Application Success

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Abstract

More unconventional oil and gas operations require low temperature dissolvable tools. The sealing element is a critical component of the dissolvable tools. The dissolvable rubbers typically used at low temperature (LT) either have low strength or the dissolution rate is low. A series of novel polyurethane dissolvable rubbers have been developed to meet the industrial requirements. A dissolvable plug with the dissolvable rubber elements has been successfully tested in the lab and used in many field cases.

Polyurethane rubbers were designed by designing a molecular structure and catalyst to balance the dissolution and mechanical properties. The mechanical properties of the dissolvable rubbers were tested at ambient and high temperatures. Dissolvable rubber coupons were soaked in 1% KCl at 40 and 50 °C for various durations to characterize the dissolution rate. In addition, a dissolvable plug with the dissolvable rubber sealing element was tested in water at 60°C, 60-70 MPa for 24 hours. The dissolution rate of the dissolvable plug was tested at 50 and 60°C in brine.

The 100% modulus of the dissolvable rubber at high temperature was higher than 600 psi and the elongation of the dissolvable rubber at high temperature was higher than 300%, which was higher than that of most commercial dissolvable rubbers. For 50-80°C rated dissolvable rubbers, the dissolvable rubber coupons were disintegrated in brine at 50°C in 3-7 days. For 40-70°C rated dissolvable rubbers, the dissolvable rubber coupons were disintegrated in brine at 40°C in 3-7 days, which is faster than that of the most commercial dissolvable rubbers. The dissolvable plug with the dissolvable rubber sealing elements held 60 MPa pressure differential at 60°C in water for more than 24 hours and 70 MPa pressure differential at 50°C. On the other hand, the dissolvable rubber element disintegrated in 50°C, 1% KCl in 7 days and 60°C, 1% KCl in 4 days. The dissolvable plugs with the low temperature dissolvable rubber sealing elements have been successfully used in many Oilfields in China, including Southwest, Changqing, Xinjiang, Zhejiang Oilfields. The low temperature dissolvable plug outperformed most dissolvable plugs in the market in terms of pressure holding and dissolution properties in the field applications. Several successful field application cases will be reported in the paper.

CNPC-USA designed a unique polyurethane rubber backbone structure, catalyst to reach an optimum balance of mechanical and dissolution properties up to 40°C or lower. The dissolution and mechanical properties of the dissolvable rubber could be designed to meet various conditions and requirements in the field. The dissolvable rubber element's manufacturing cost is very competitive. The low temperature dissolvable rubber that we developed could have many applications in other downhole tools and other industries.

Introduction

Hydraulic fracturing of horizontal wellbores in tight oil and tight gas formations is a common practice to exploit unconventional resources. Plug and perf technology is the most commonly used method in the industry. Right now, both polymer based composite plugs and dissolvable plugs including dissolvable metal-based plugs, and dissolvable Polyglycolic acid (PGA) based plugs have been used. The dissolvable plugs have also been used extensively in long reach horizontal wells. The water dissolvable elastomer sealing element is an essential component of the dissolvable plugs. The dissolvable elastomer needs to keep good elasticity for certain times, such as more than 12 hours to complete the fracturing operation. On the other hand, the dissolvable elastomer needs to be dissolved as fast as possible after performing the fracturing operation. It is very challenging to develop the dissolvable rubbers to satisfy both requirements.

The polymer backbone for the traditional rubbers, such as natural rubbers, nitrile rubber, fluororubbers, etc., are not water soluble. Even though the dissolvable plug has been used in the industry for more than 15 years, there are still limited publications on the development of dissolvable rubbers. Scientists tried different methods to develop dissolvable rubbers. Some scientists compounded traditional rubber with dissolvable polymers, such as PGA, polyurethane. However, it is quite challenging to control the traditional rubber as an isolated phase in the dissolvable polymer matrix after vulcanization. Some researchers tried thermoplastic polyurethane. However, the strength of the elastomer is low. In the past, the dissolvable polyurethane rubbers have been developed to meet the medium temperature application in the range of 80-120°C (Ren, et al., 2021) and high temperature application at 150°C (Yue, et al., 2022). Special rubber and coatings have been developed for the acid fracturing applications at 120-150°C (Ren, et al., 2024).

The Permian Basin in West Texas produces approximately 2 million b/d and accounts for over 20% of U.S. crude production. (Walton, et al., 2017). The formation temperature is low in the Permian basin and some other U.S. shale gas reservoirs, such as Marcellus, Fayetteville. The average downhole temperature for these formations ranges from 100 to 150°F (38 to 66°C). The dissolution process slows in lower-temperature shale formations, and the elastomer often ceases degradation at these lower temperatures. To meet the requirements of dissolvable rubber for low temperature applications, in the past various low temperature rubbers have been reported for the temperature up to 125°F (50°C) (Ping, et al., 2018, Walton et al., 2017). A series of novel advanced thermoset polyurethane based dissolvable rubber materials were developed at CNPC to target applications up to 40°C. The dissolvable rubbers display a fast dissolution rate and high mechanical strength. The dissolvable plug using the low temperature dissolvable rubber elements have been tested in the lab to hold pressure differential of 70MPa at 50°C and 60°C for 24 hours. The dissolvable plug could be fully dissolved in 7 days. The dissolvable plugs have been successfully used in many oil fields in China.

Materials and Methods

There are multi-functional groups in the backbone of the polyurethane dissolvable rubber polymers including ester group, urethane group, urea group. The hydrolysis reaction of multi-function groups makes the dissolvable rubber dissolve faster. Special catalysts were designed to further accelerate the dissolution rate of the dissolvable rubbers. In addition, the dissolvable polyurethane elastomer is a thermoset polymer, which makes the elastomer have good elasticity and mechanical strength. The dissolution and mechanical

properties of the materials could be adjusted by designing the molecular structure and crosslinking agent. Dissolvable rubbers DR-044, DR-045, DR-046 were developed for 50-80°C applications and the hardness (Shore A) of these dissolvable rubbers are 85. DR-067, DR-068 were in developed for 40-70°C applications and the hardness (Shore A) of these dissolvable rubbers are in the range of 75-80.

The tensile properties of the dissolvable rubbers were tested at ambient and 40, 50°C based on ASTM D412 testing standard. The tensile testing coupons and round coupons were soaked in 1% KCl at 40 and 50 °C for various durations up to 21 days. The mass, dimensions, volume, density, hardness of the dissolvable rubbers, pH value of the fluid were measured at various duration to characterize the dissolution rate.

In addition, the pressure holding test of the various dissolvable plugs with the low temperature dissolvable rubber sealing element was tested in water at 60°C, 60 MPa for 24 hours. The dissolution rate of the dissolvable plug was tested at 50°C or 60°C in brine. The weight loss, hardness change, dimension of the residues, and pH value of the fluids were measured to monitor the dissolution progress of the dissolvable plugs and dissolvable sealing elements.

Results and Discussions

Properties of Dissolvable Rubbers

50-80°C dissolvable rubbers. Fig. 1 displays the dissolution progress of the three materials soaked in 1%KCl at 50°C. All three materials were intact after soaking in water for 8 hours, the samples were cracked on the surface within one day. The water disintegrated into fine particles within 3-7 days, demonstrating excellent low temperature dissolution properties. The cracking of the dissolvable rubbers could be due to the diffusion process of the water to the core of the dissolvable rubber coupons and disintegrated the rubber molecular structure. DR-045's hardness decreased by 38% within 24 hours and its weight reduced by 56% within four days. The weight reduction rate of DR-045 at 50°C is comparable to that of medium temperature dissolvable rubber DR-048 at 80°C. (Ren, et al., 2021) A sharp reduction of the weight of DR-044, DR-045, DR-046 were observed when soaked in the brine for 10 days, which is likely due to disintegration of the samples to tiny pieces. The weight of DR-044, DR-045, DR-046 reduced 58%.80%, 85% after 10 days, respectively. The results suggest the low temperature dissolvable rubber element's dissolution rate is high.

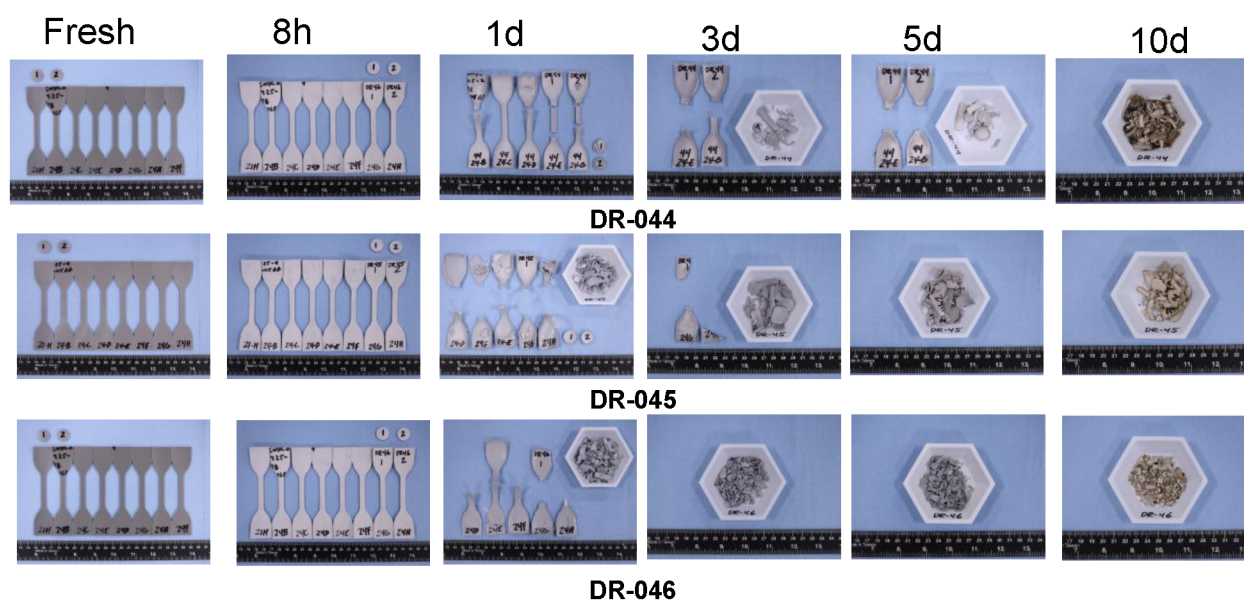


Figure 1—Dissolution progress of DR-044, DR-045, DR-046 at 50°C, 1%KCl

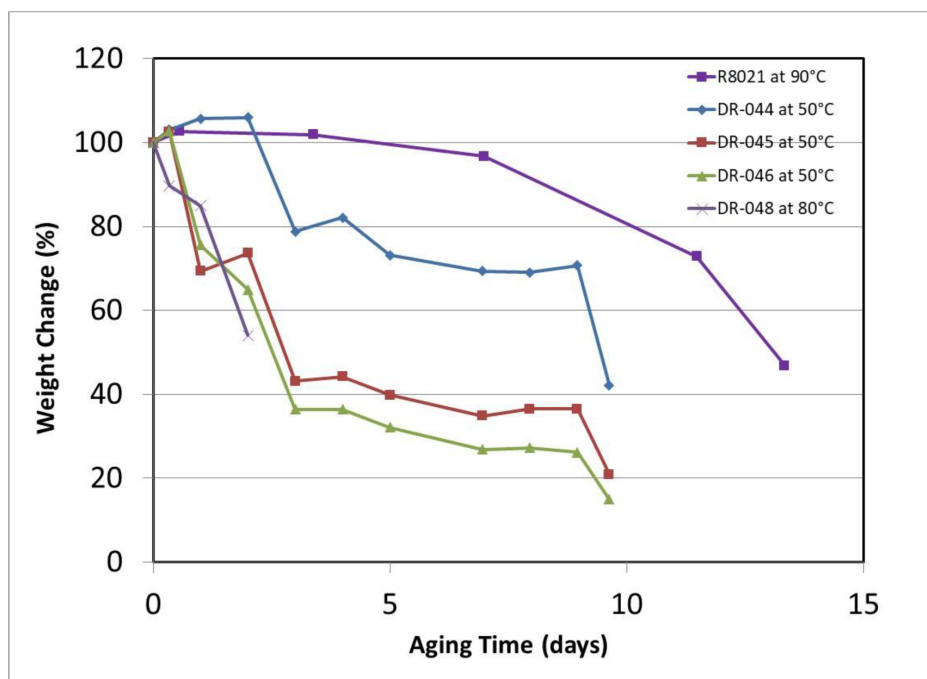


Figure 2—Weight change of DR-044, DR-045, DR-046 at 50°C, 1%KCl

Fig. 3 displays the stress-strain curve of DR-044, DR-045, DR-046 at 50°C. The mechanical behavior of the three materials is similar. The 100% modulus of DR-045 at 50°C is about 630 psi and the elongation is above 650%. The 100% modulus of DR-045 at 50°C is comparable to the medium temperature dissolvable rubber R8021 at 90°C (Ren, et al., 2021), which suggests the dissolvable rubber element could hold pressure differential of 70 MPa at 50°C. Both DR-045 and DR-046 display high dissolution rate, high mechanical strength, while DR-045 is more economical for processing. DR-045 was selected for further dissolvable rubber sealing element testing.

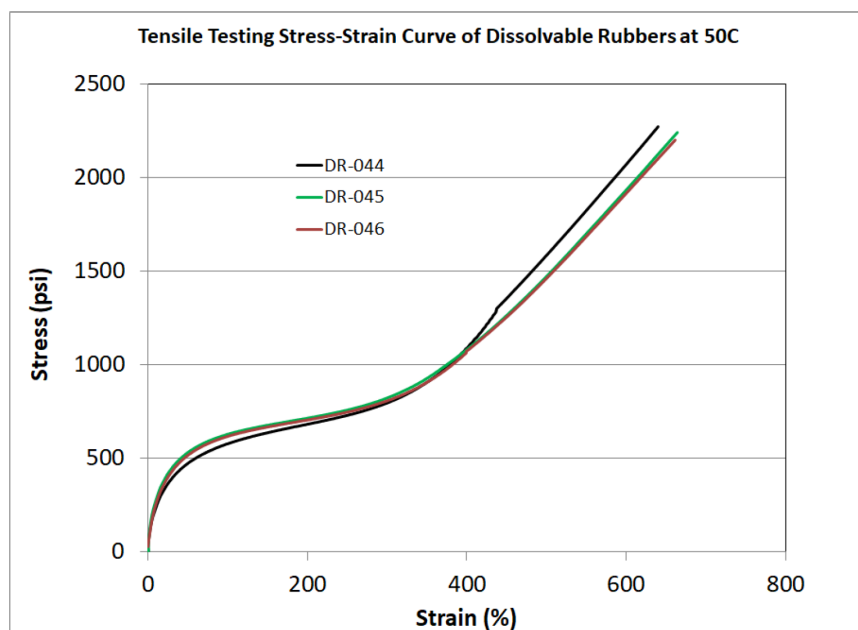


Figure 3—Stress-strain curve of DR-044, DR-045, DR-046 at 50°C

40-70°C dissolvable rubbers. Fig. 4 shows the dissolution progress of DR-067, DR-068 at 40°C in 1%KCl. Both materials were intact after soaking for 8 hours. DR-067 and DR-068 broke apart in the solution in 1 and 3 days, respectively. DR-068 broke into small pieces in 9 days. Fig. 5 shows the weight reduction of DR-068 and DR-067 is higher during the first 5 days, while the weight reduction rate is slow afterwards. After 5 days, the weight loss of DR-067 and DR-068 was 55% and 62%, respectively. After 17 days, the weight loss of DR-067 and DR-068 was 64% and 68%, respectively. The weight reduction rate of DR-068 at 40°C is comparable to that of DR-045 at 50°C, which suggested that the dissolution rate of DR-068 sealing element at 40°C could be fast too. Both DR-067 and DR-068 have high elongation at break at 40°C (>650%). The 100% modulus of the two materials at 40°C is a little lower than that of DR-045 at 50°C. With proper sealing element and back-up design, the sealing element made of DR-068 could hold high pressure differential based on the previous experience.

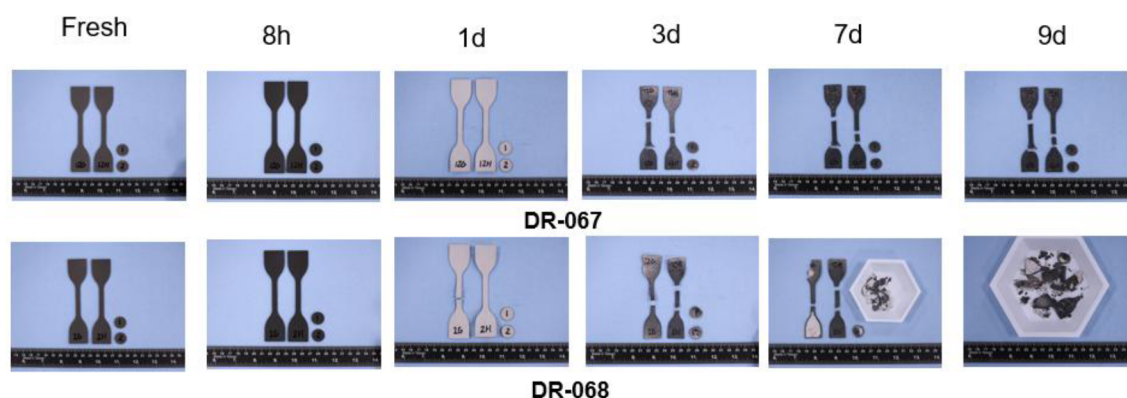


Figure 4—Dissolution progress of DR-067, DR-068 at 40°C, 1%KCl

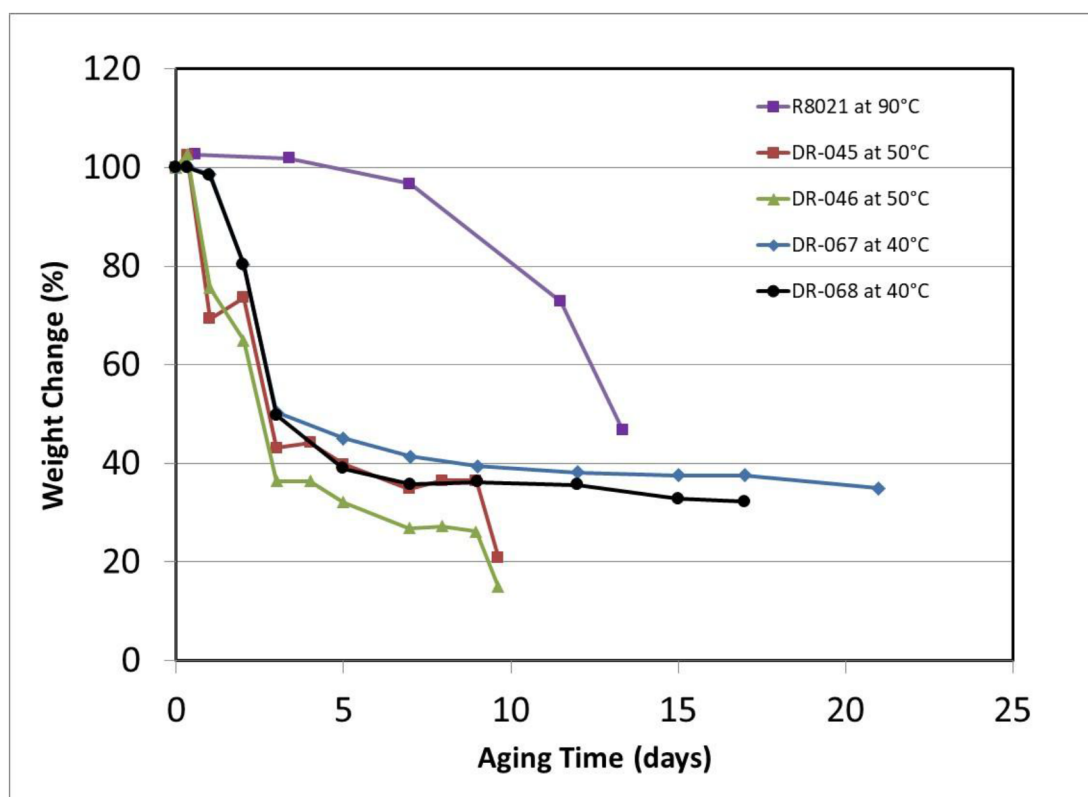


Figure 5—Weight change of DR-067, DR-068 at 40°C, 1%KCl

Table 1—Tensile testing results of DR-067, DR-068 at 40°C

Materials	50% Mod (psi)	100% Mod (psi)	200% Mod (psi)	300% Mod (psi)	Tensile Strength (psi)	Elong. at Break (%)
DR-067	335	420	526	659	>1900	>650
DR-068	341	428	526	652	>1900	>650

Properties of Dissolvable Plug with LT Dissolvable Rubber Sealing Element

A 101 mm outside diameter (OD) dissolvable rubber sealing element based on DR-045 was manufactured for dissolution testing. The element was soaked in 1% KCl at 50°C. The dissolvable rubber element kept in good shape after soaking in the fluid for 1 day. The surface of the element start to crack after soaking in the fluid for 2 days, the element broke apart in 3 days and broke to tiny pieces with the maximum dimension of the residues less than 2 cm in 7 days. The weight reduction of the element is more than 60% after soaking for 7 days, which is consistent with the coupon dissolution testing results. The hardness of the residue reduced 60% after aging for 4 days. The residues have no strength and are easy to flow back to surface.

A 101 mm dissolvable plug made of DR-045 element was designed at CNPC-USA. The pressure holding test of the plug was performed in water at 60°C (140°F). The plug was set in 5.5", 23 lb/ft casing. The setting force was 29,110 lbs. After the setting, the plug passed 8,700psi pressure holding for 10 minutes at ambient temperature. Afterwards, the temperature increased to 60°C (140°F). During the 24-hour holding period, the upper zone lost 600 psi. The plug did not move downwards during the post inspection. The plug successfully passed 8,700psi pressure holding at 60°C for 24 hours (Fig. 7).

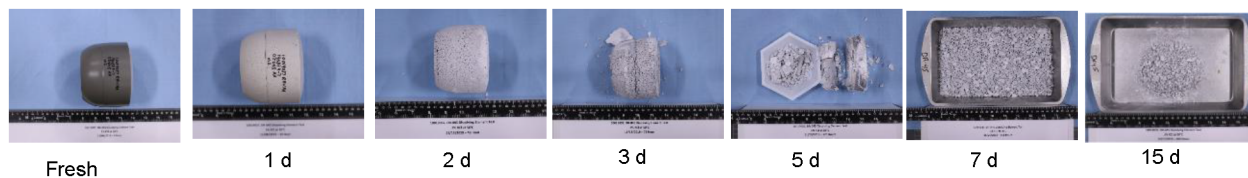


Figure 6—Dissolution progress of DR-045 element at 50°C in 1%KCl

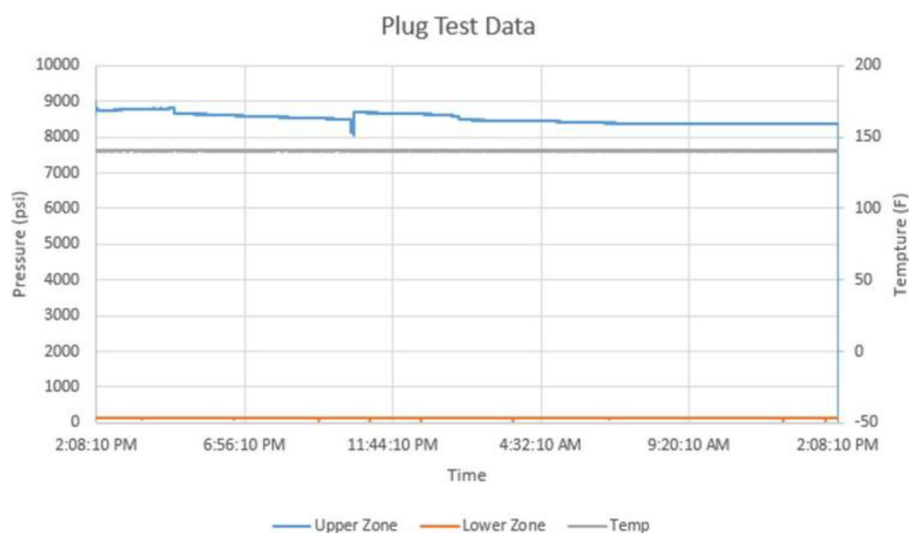


Figure 7—Pressure holding test of 101 mm dissolvable plug with DR-045 sealing element at 60Mpa, 60°C for 24 hours

After the dissolvable plug pressure holding testing, the 101 mm dissolvable plug was soaked in 1% KCl at 60°C for dissolution testing. The dissolvable rubber element broke into pieces after 12 hours. All the dissolvable metal parts were dissolved within 24 hours. The dissolvable rubber element broke into tiny pieces with just four pieces larger than 2 cm after 72 hours. After 4 days, the whole dissolvable rubber element broke into small pieces less than 2 cm. The total weight of the dissolvable plug reduced 95% in 5 days. (Fig. 9) Various designs of LT dissolvable plugs with LT dissolvable rubber sealing elements have successfully passed pressure testing in water at 70 MPa at 50 and 60°C.

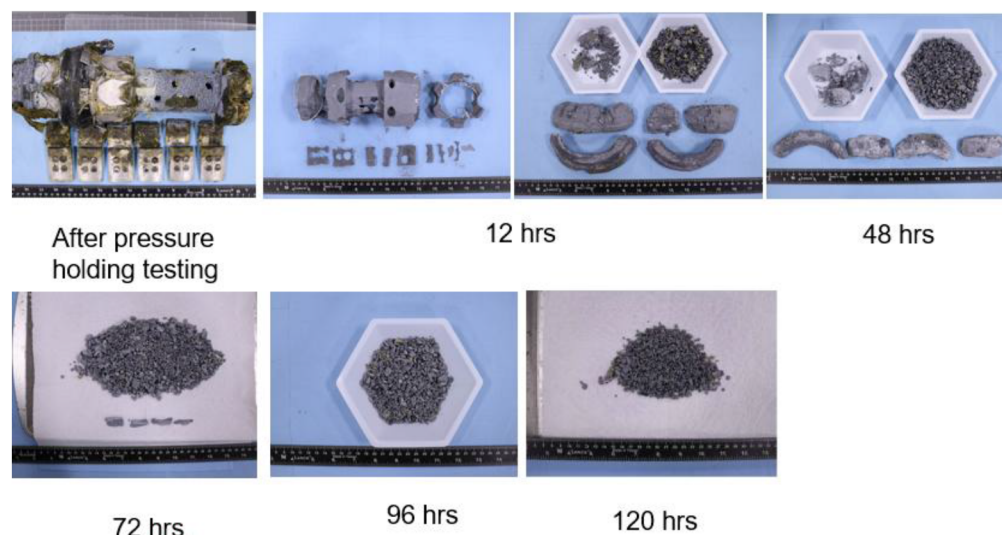


Figure 8—Dissolution progress of 101 mm dissolvable plug with DR-045 sealing element at 60°C, 1%KCl

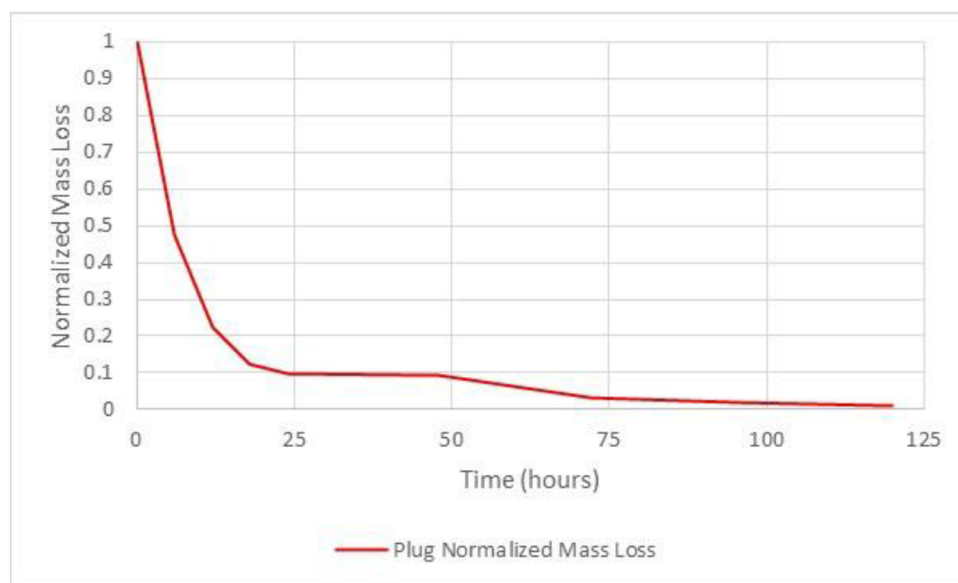


Figure 9—Weight reduction of 101 mm dissolvable plug with DR-045 sealing element at 60°C, 1%KCl

Successful Field Cases

Field Case #1. The low-temperature dissolvable plugs with dissolvable rubber sealing element DR-045 were scaled up for use in the horizontal well fracturing operations in shallow shale gas formation in the Sichuan-Chongqing region. A total of more than 50 horizontal wells and over 800 stages have been completed. Before fracturing process, operations such as pumping plugs, setting, dropping dissolvable balls

were successfully carried out, providing effective support for smooth multi-stage fracturing and reservoir stimulation.

The horizontal wells in this region have a vertical depth of 800m to 1500m, a total depth of 2300m to 3590m, a horizontal section length of 905m to 1700m, a downhole temperature range of 35°C to 52°C, downhole salinity ≤ 1000 ppm, and a maximum surface working pressure of ≤ 65 MPa. The casing outer diameter is 139.7mm (5.5in), inner diameter 118.62mm (23 lb/ft) or 121.36mm (20 lb/ft), and steel grade P110. The 98, 103, 110mm OD low-temperature dissolvable plugs, developed based on the specific operating conditions of this region, have achieved the optimal fracturing, ensuring smooth wellbore access without obstruction. Fig. 10 displays a typical set of curves for fracturing operations for the region.

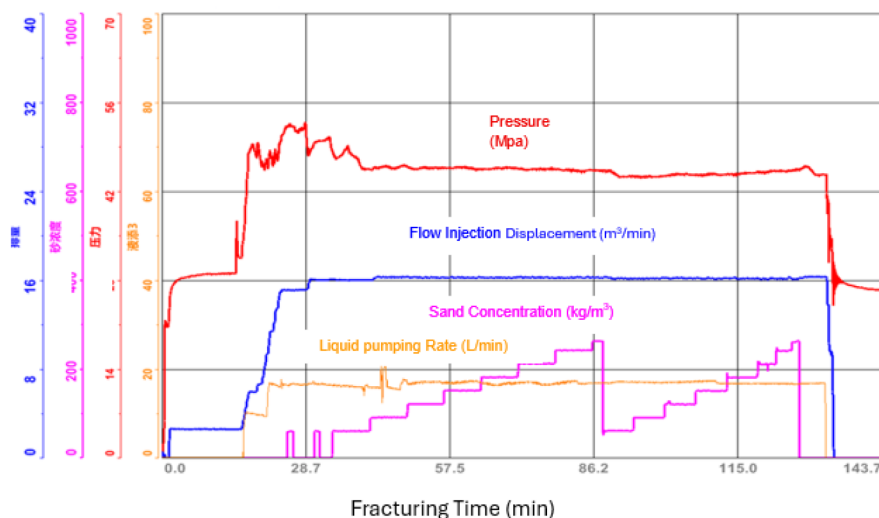


Figure 10—Fracturing operation curves for YS137H17-9 well #5 stage in Sichuan Province of China

After fracturing and the well was shut-in, the dissolvable plugs dissolved autonomously in the wellbore formation fluid, coil tubing with milling tool was used to test the obstruction. The wells with the low temperature dissolvable plug did not have obstruction. The flow back residue is normally less than 0.4 kg ensuring smooth wellbore access without obstruction and enabling quick production from the oil and gas wells. This solves the issue of hard milling problems for dissolvable plugs and dissolvable rubber sealing elements after fracturing, significantly reducing production risks, shortening the production cycle, and lowering operational costs. Through large-scale application in the horizontal well fracturing operations in this region, the comprehensive performance of the dissolvable plugs in terms of pressure holding during fracturing and fast dissolution properties has been fully validated, meeting the reliability and stability requirements for operations under complex conditions.

Field Case #2. For the ultra-low temperature reservoirs of Changqing tight oil, conventional dissolvable plugs with OD 111 mm and ultra-short, ultra-large diameter dissolvable plugs with OD 117 mm were developed. Additionally, for the 4.5-inch casing Changqing tight gas well, an ultra-short dissolvable plug with OD 90 mm was developed. All the three types of dissolvable plugs have passed 70MPa pressure holding test for 24 hours. Two types of dissolvable plugs were carried out field testing for Changqing Oilfield tight oil and tight gas formation, respectively. The dissolvable plugs were tested in 5 wells. The plug setting and pressure holding performance were stable. The fracturing operation curves were smooth, and the overall performance of these dissolvable plugs met the well completion and fracturing requirements for Changqing Oilfield tight oil and tight gas formations.



Figure 11—Low temperature dissolvable plug field application in Changqing oilfield of China

Conclusions

A unique thermoset polyurethane low temperature dissolvable rubber was developed by designing backbone molecular structure, crosslinking agent. Special catalysts were developed to accelerate the dissolution rate of the dissolvable rubbers. With the design of the molecular structure and catalysts, an optimum balance of mechanical and dissolution properties was achieved. Two categories of LT dissolvable rubbers were developed for 50-80°C and 40-70°C applications. Compared with the low temperature dissolvable rubbers in the market, the low temperature dissolvable rubber developed at CNPC-USA has the following advantages:

1. Higher strength, higher elongation, higher tear strength
2. Faster dissolution rate, easy to flow back to surface.
3. Wider temperature rating

Based on our best knowledge, this is the first report for dissolvable rubber for 40°C applications. Various sizes (110mm, 103 mm, 98mm) and design of dissolvable plugs with the dissolvable rubber sealing elements were developed. The dissolvable plugs could hold pressure either at 50°C or 60°C in water for 24 hours up to 70 MPa. The dissolvable rubber element dissolved in brine at 60°C in 4 days and 50°C in 7 days. The low temperature dissolvable plugs have been used extensively in the oil fields, such as Southwest, Changqing, Zhejiang oilfields in China. Excellent feedback was received from oil fields: The dissolvable plug holds fracturing pressure well and have good dissolution properties. The dissolvable rubber element's manufacturing cost is very competitive compared with the market price of the dissolvable rubber element in the market. The low temperature dissolvable rubber that we developed could have many applications in other downhole tools and other industries.

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